

JavaScript Ultimate Introduction Notes

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■ 1. What is JavaScript?

JavaScript is a programming language used to make websites interactive and dynamic. It adds logic to webpages.

HTML = Structure **CSS** = Design **JavaScript** = Action

■■■ 2. Who Developed JavaScript?

JavaScript was created by **Brendan Eich** in 1995 at Netscape. It was built in about 10 days.

■ 3. Story Behind JavaScript

Earlier websites were static. Users had to reload pages for every action. Netscape wanted websites to respond instantly, so JavaScript was introduced.

■■ 4. Why the Name JavaScript?

The language was first called **Mocha**, then **LiveScript**, and finally **JavaScript**. Java was popular then, so the name helped marketing.

■ 5. Browser Evolution

WorldWideWeb → Netscape Navigator → Internet Explorer → Firefox → Chrome → Edge.

■■ 6. Where JavaScript Runs?

JavaScript runs inside browsers such as Chrome, Firefox, Edge, and Safari. It also runs outside the browser using **Node.js**.

■ 7. Why Console?

The browser console helps developers test code, print output, and debug errors quickly.
Open using: Right Click → Inspect → Console.

■■ 8. Ways to Execute JavaScript

1. Inline Script 2. Internal Script 3. External JS File 4. Browser Console

■ 9. JavaScript Engines

Chrome uses **V8**, Firefox uses **SpiderMonkey**, and Safari uses **JavaScriptCore**.

■ 10. How JavaScript Executes

JavaScript is a **High Level Language**. Engines convert readable code into machine code for the computer.

■ 11. Compiler / Interpreter / JIT

Compiler converts code before execution. Interpreter runs code line by line. Modern JavaScript uses **JIT (Just In Time)** for better speed.

■ 12. Install Node.js

Download the LTS version from the official Node.js website. After installation, check with **node -v**.

■■ 13. Run JS in Node.js

Create a file named **app.js** and write:

```
console.log('Hello');
```

Then run:

```
node app.js
```

■ 14. Why console.log() ?

console.log() is used to print values, messages, and check program flow during learning and debugging.

■ 15. Example Program

```
a = 100;  
console.log(a);  
console.log(typeof a);
```

Output: **100, number.**

■ 16. Data Types

Primitive: number, string, boolean, undefined, null, bigint, symbol.

Non-Primitive: object, array, function.

■ 17. var, let, const

var is old style. **let** is used when value changes. **const** is used when value should stay fixed.

Best practice: Use **const** by default.

■ 18. Next Topics

Operators • Conditions • Loops • Functions • Arrays • DOM • Events • ES6

■ Quick Comparison Table

Keyword	Redeclare	Reassign	Scope	Best Use
var	Yes	Yes	Function	Avoid
let	No	Yes	Block	When value changes
const	No	No*	Block	Default choice

■ Learn daily, code daily, grow daily.